Data Analysis and Visualization
PROJECT

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Roll No : 59
Batch : B4
Date: 06/04/2026
Aim: Design an interactive dashboard using Power BI.

Introduction

This project focuses on the analysis of Indian Premier League (IPL) data using Microsoft Power BI, a powerful business intelligence and data visualization tool. The aim of this project is to transform raw cricket data into meaningful insights that help in understanding patterns related to team performance, player contributions, and match outcomes.

The dashboard provides an interactive platform where users can explore key aspects such as match results, scoring trends, player statistics, and the impact of strategic decisions like winning the toss. By leveraging data visualization techniques, the project enables better interpretation of complex datasets across multiple IPL seasons.

This analysis not only enhances data-driven decision-making but also demonstrates the practical application of data analytics in the sports domain.

Objectives

- Analyze team performance across seasons
- Identify top batsmen and bowlers
- Study match trends and scoring patterns
- Evaluate toss impact on match results
- Visualize data using interactive dashboards

Dataset Description

The dataset used contains IPL match and delivery data including:

- Matches dataset: match details, teams, winners, venue, toss
- Deliveries dataset: ball-by-ball data including runs, wickets, players

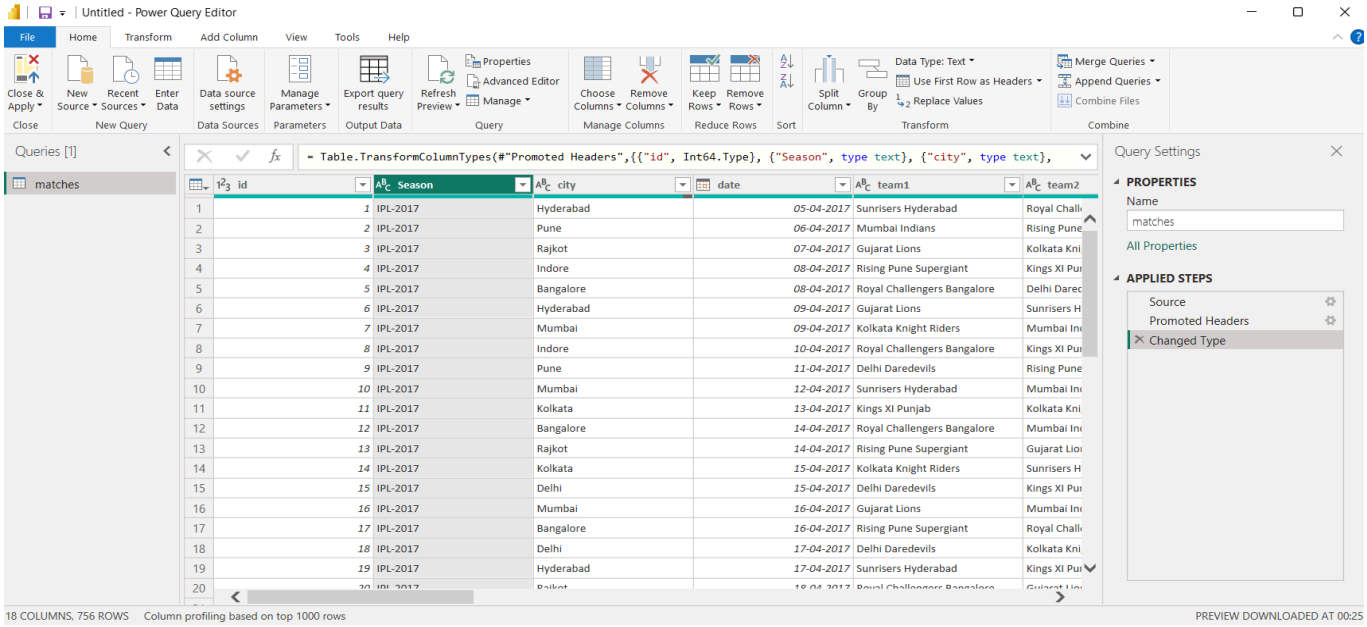
Source: Kaggle IPL Dataset

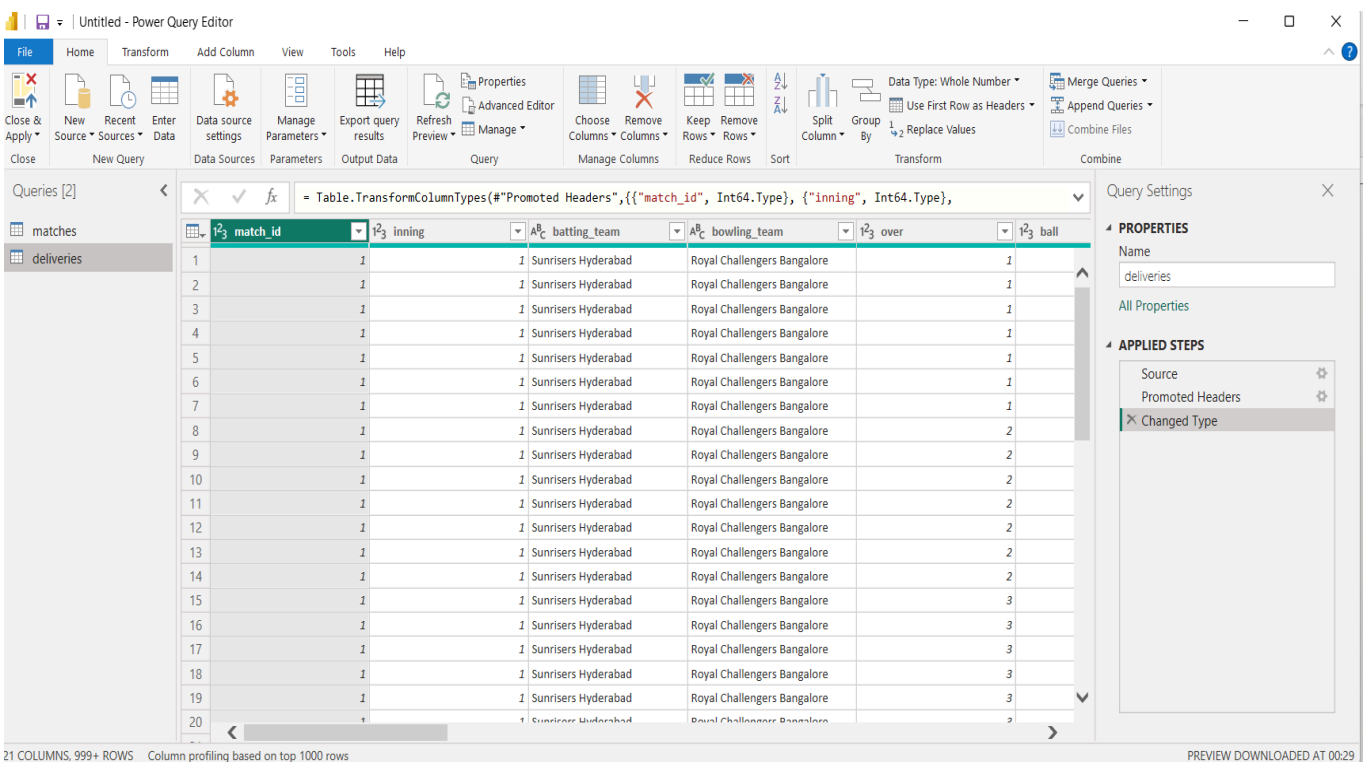
Tools Used

- Microsoft Power BI Desktop
- CSV Dataset (IPL Data)

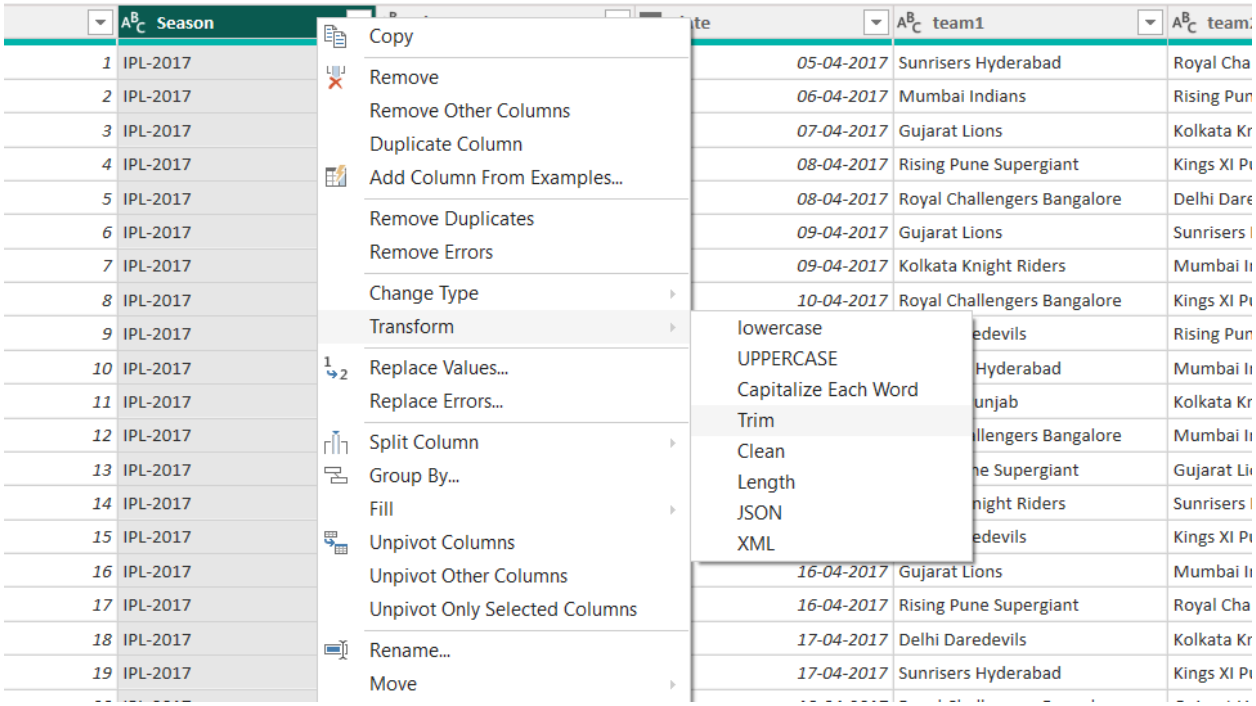
Steps Performed

1. Imported IPL dataset into Power BI

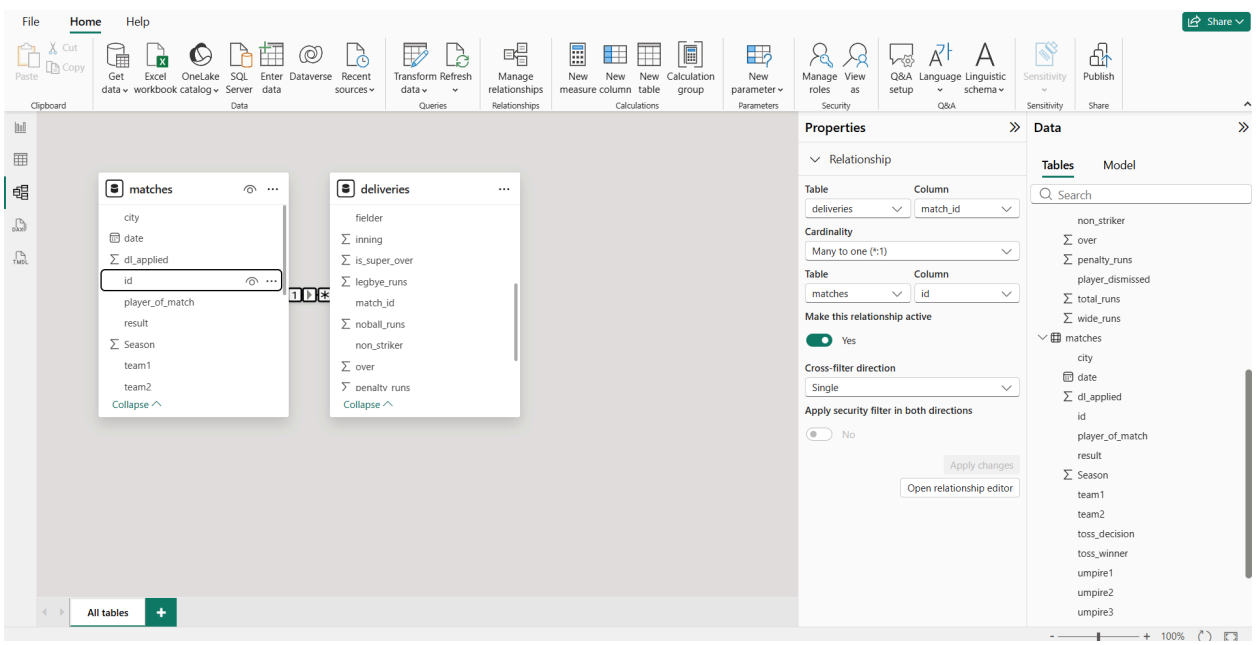




2. Performed data cleaning and transformation using Power Query



3. Established relationships between tables (matches & deliveries)



4. Created calculated columns and measures

1. **Avg Runs per Match = DIVIDE(SUM(deliveries[total_runs]), COUNT(matches[id]))**
2. **[Total Matches] [Total Runs] [Avg Runs] [Total Wickets]**
3. **Toss Impact =**

```
IF(
    SELECTEDVALUE(matches[toss_winner]) = SELECTEDVALUE(matches[winner]),
    "Won Toss & Match",
    "Lost After Toss"
)
```

4. **Toss Result =**

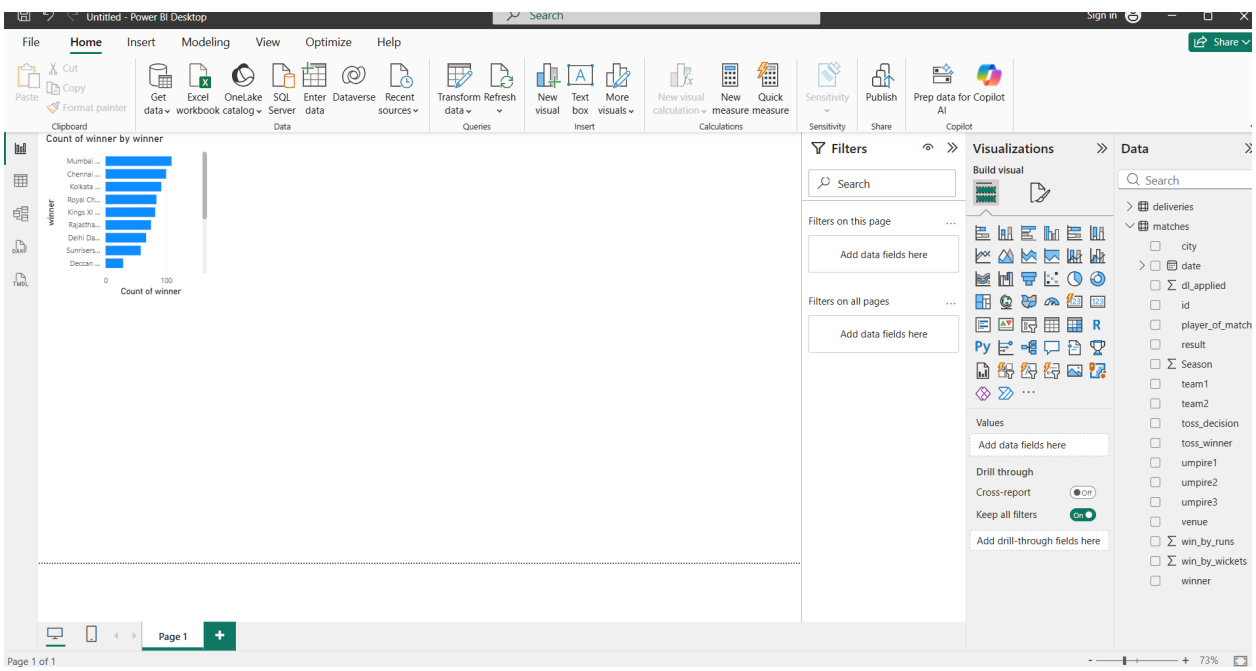
```
IF(
    matches[toss_winner] = matches[winner],
    "Toss Advantage",
    "No Advantage"
)
```

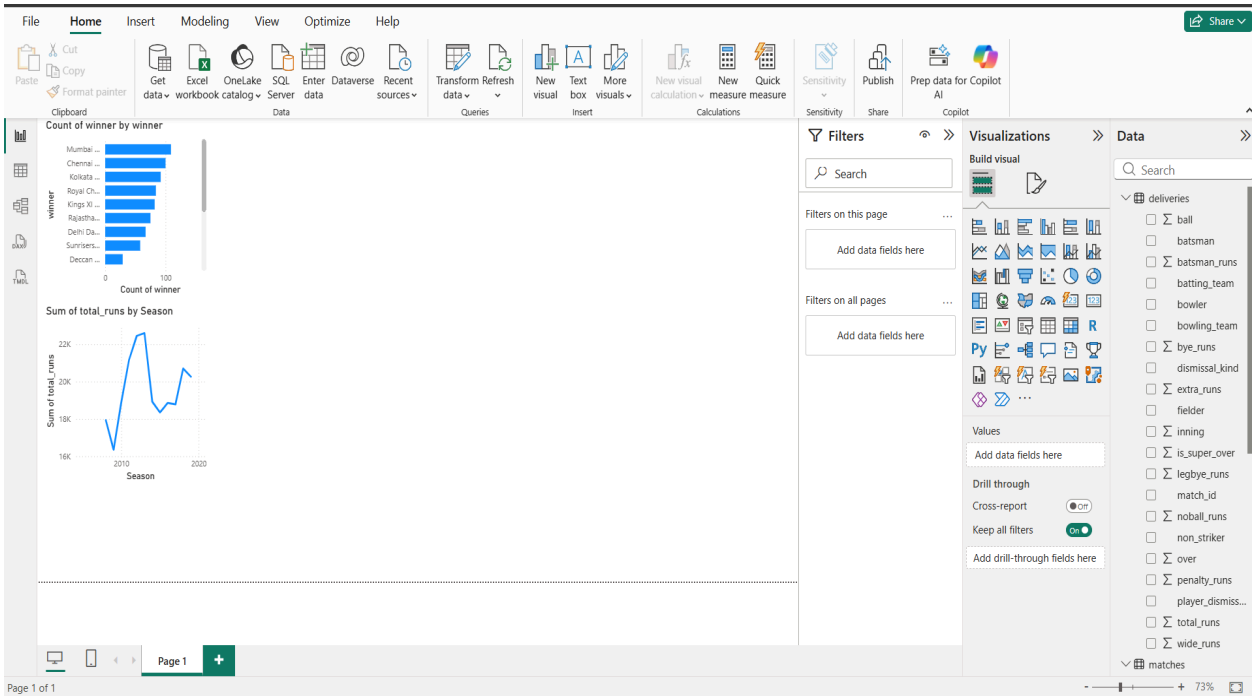
5. **Over Phase =**

```
IF(deliveries[over] <= 6, "Powerplay",
IF(deliveries[over] <= 15, "Middle",
"Death"))
```

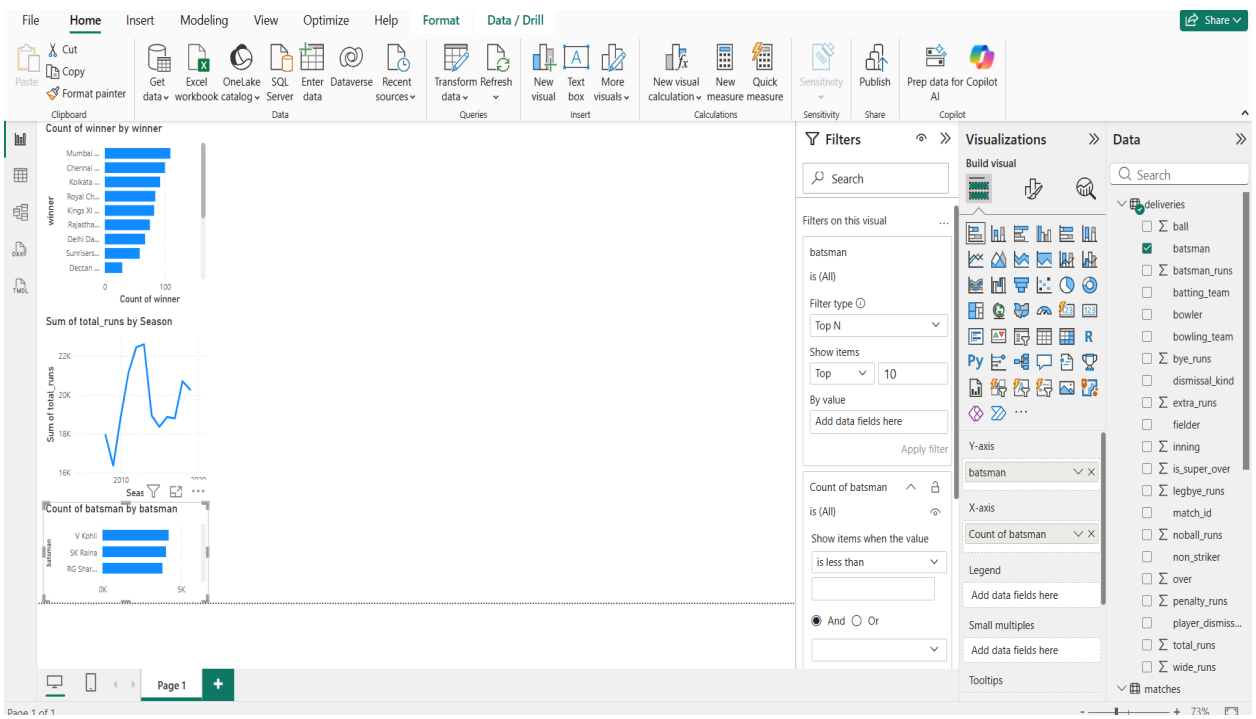
5. Built interactive visualizations:

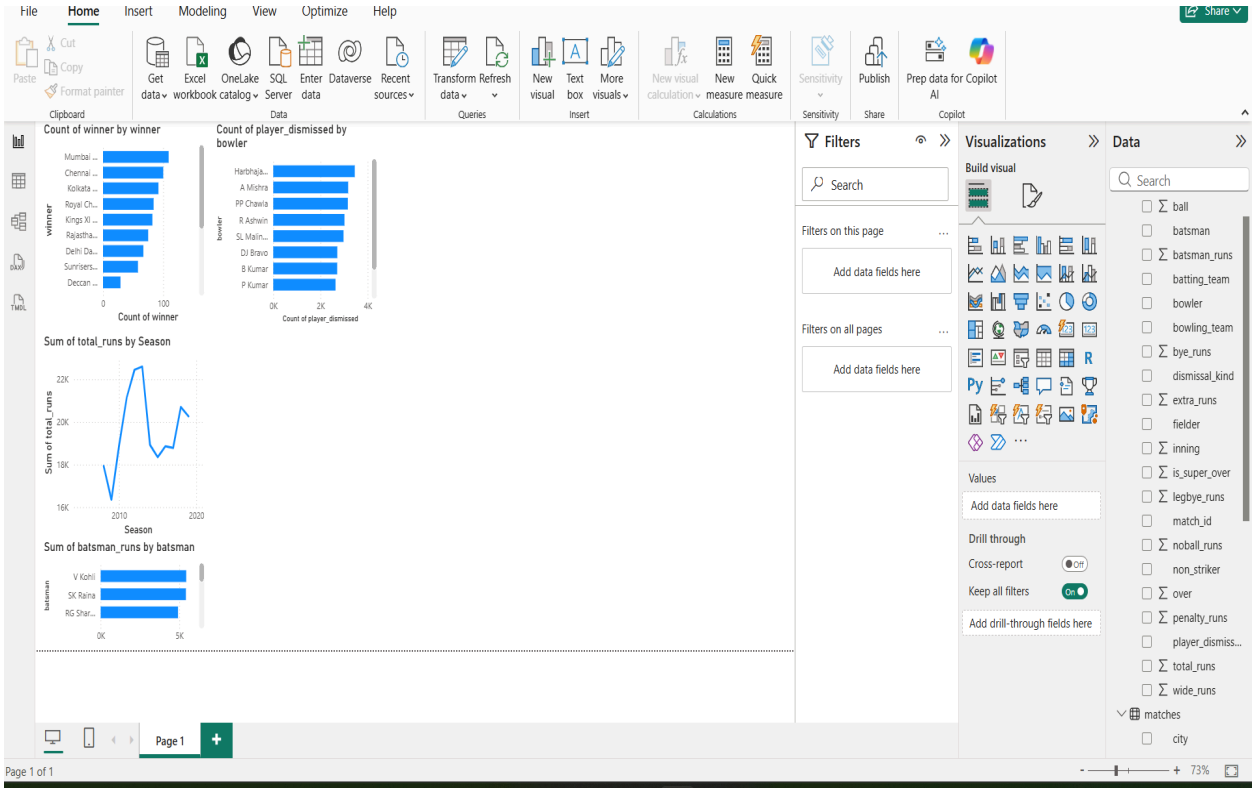
- Team performance charts



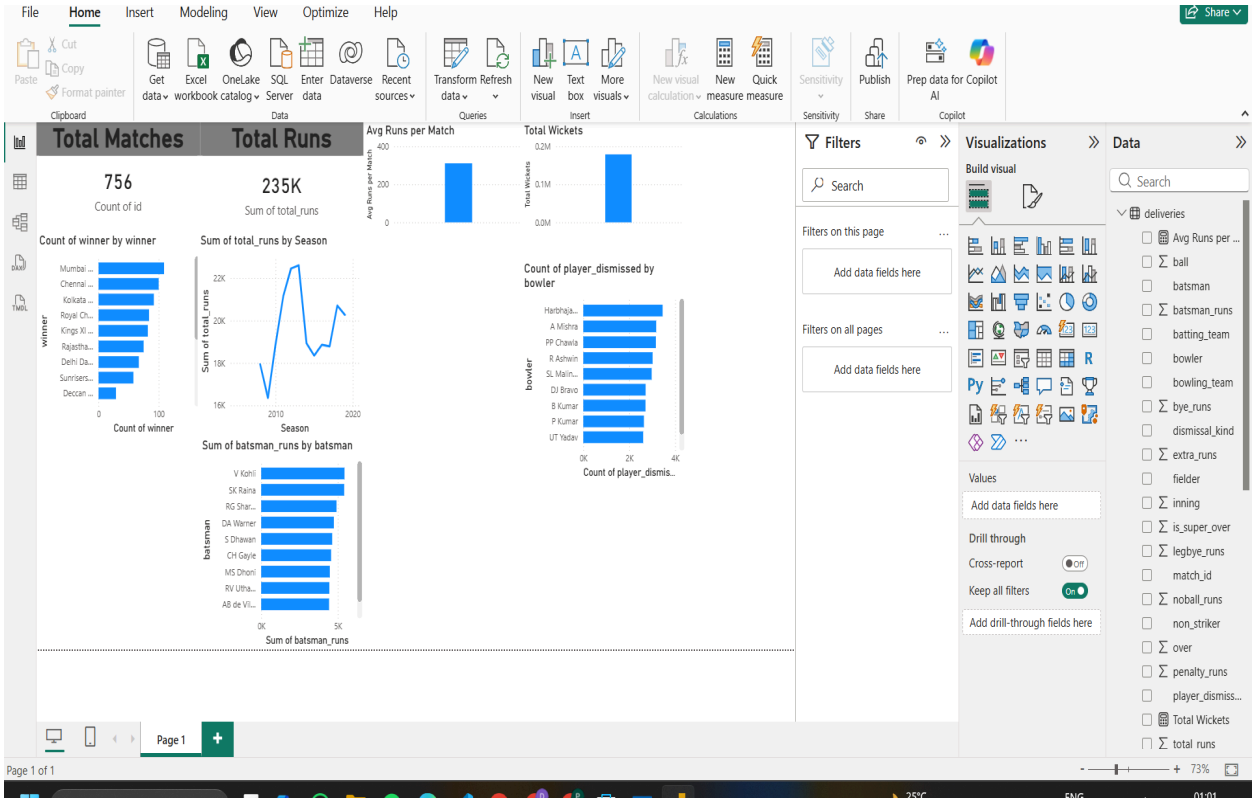


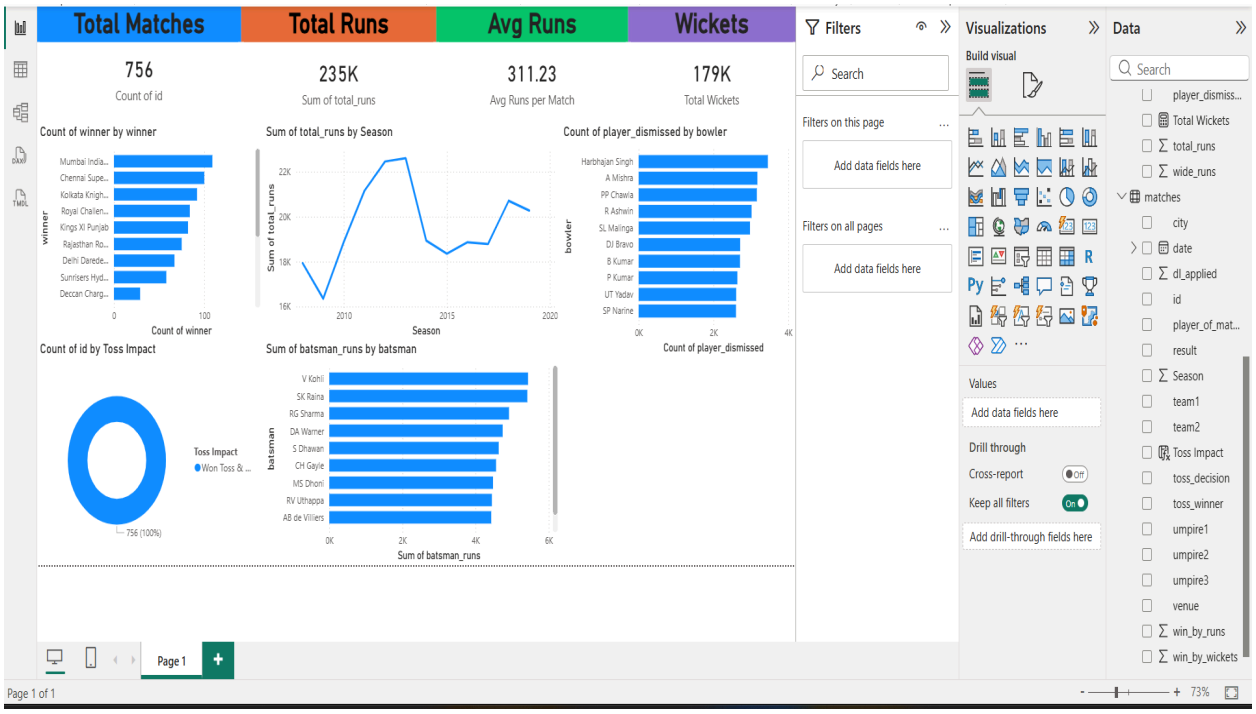
- Player performance charts



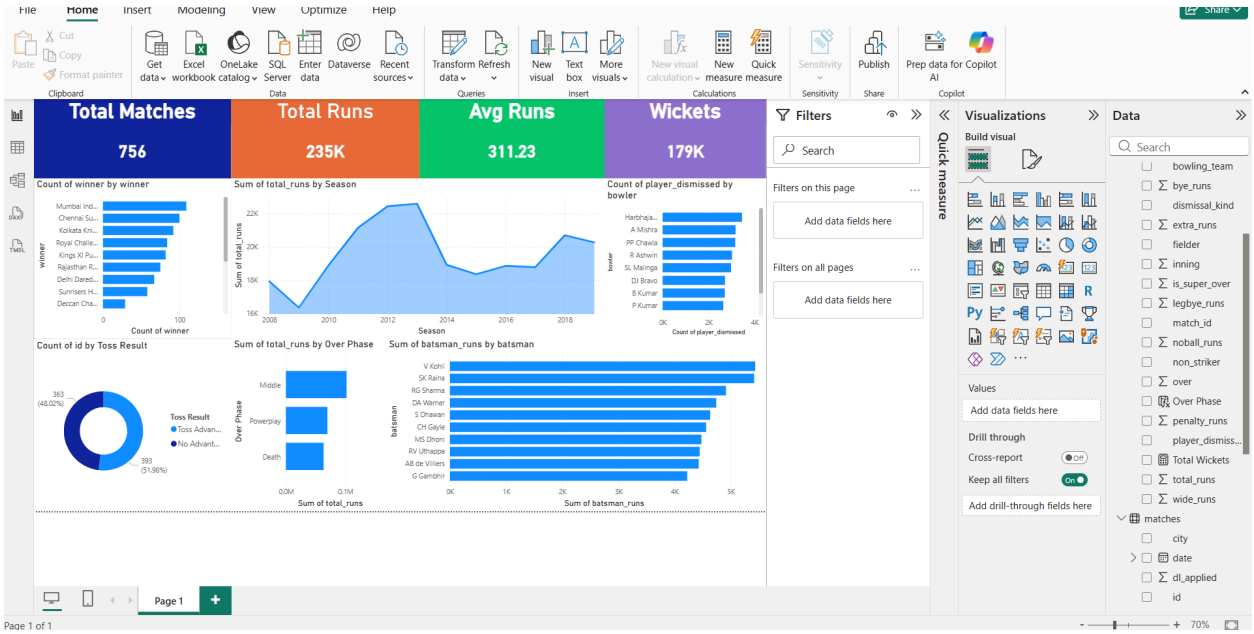


- KPI cards (Total Matches, Runs, Avg Runs, Wickets)

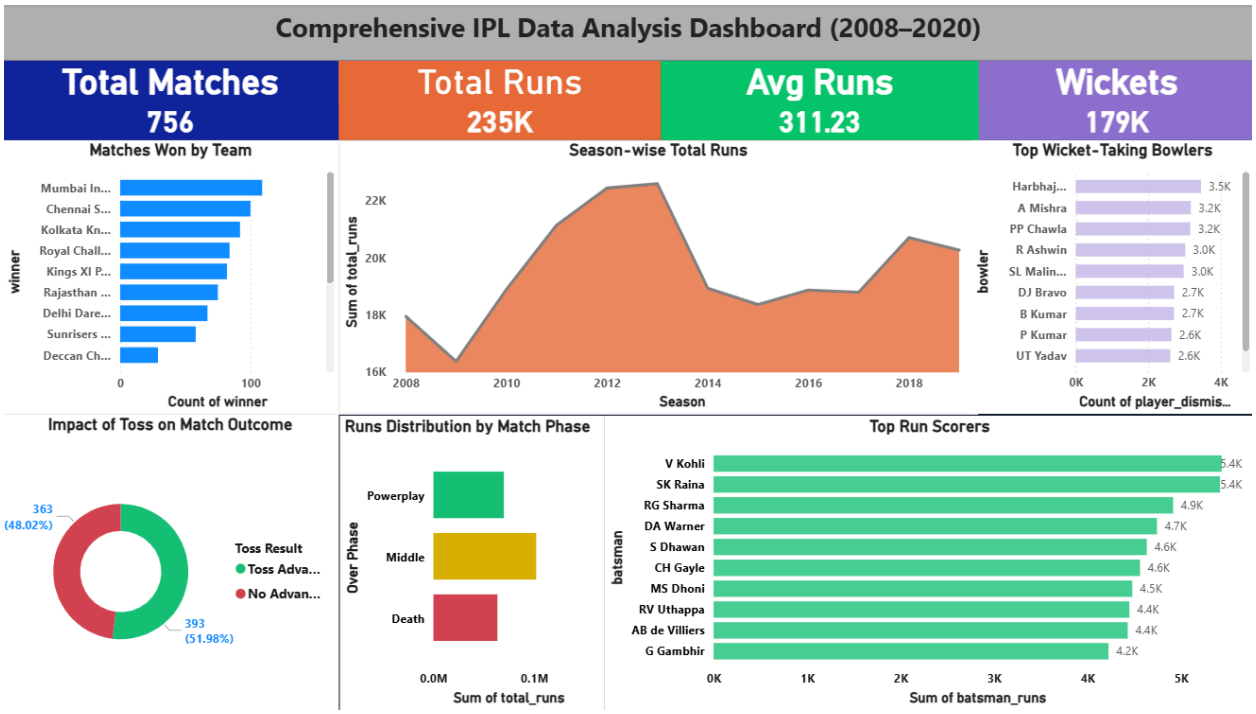




- Toss impact analysis & Over phase analysis



6. Designed and formatted dashboard



Dashboard Features

- KPI Cards showing key metrics
- Team-wise match wins
- Season-wise run trends
- Top batsmen and bowlers
- Toss impact analysis
- Over phase scoring distribution

Key Insights

- Mumbai Indians is the most successful team
- Run scoring trends vary across seasons
- Top players like Virat Kohli dominate batting
- Death overs contribute significant runs
- Toss impact is moderate (~50%)

Conclusion

The project successfully demonstrates how data visualization tools like Power BI can be used to analyze large datasets and extract meaningful insights. The dashboard provides a clear and interactive representation of IPL data.